

Haemodialysis

Fistula Needles

Product Range



**FRESENIUS
MEDICAL CARE**

Benefits of Fresenius Medical Care Fistula Needles

Fistula needles are the crucial link between the patient and the dialysis machine, requiring quality, safety and comfort for patient and user.

Biocompatibility

- All Fresenius Medical Care needles are coated with silicone for easier, smoother puncturing and to reduce blood-material interactions.

Improved geometry and flow

- Vessel-trauma and pain perception during puncture are reduced due to the good ratio of cutting and stretching, the needle's sharp tip and rounded and polished trailing edge.
- Ultra-thin walls of the needles and larger inner lumen diameters permit maximum blood flow rates.
- Lesions or limitations in blood flow are reduced as the special slit-formed back-eye of the arterial and single needle reduces suction of the needle towards the inner wall of the vessel.

Ergonomic wing design

- The convenient rotating wings enable user-friendly handling and adaptation to puncture technique. (Figure 2)
- Textured wings provide a secure grip.

Colour-coded application guidance

- Colour-coded clamps and wings enable easy differentiation of the needles. (Figures 4 & 5)

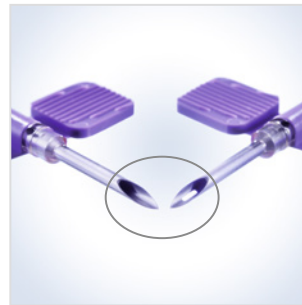


Figure 1: Special slit-formed back-eye of arterial and single needle



Figure 2: Rotating wing design allows good control and easy gripping during cannulation

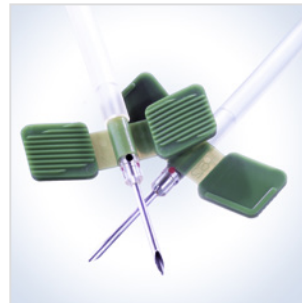


Figure 3: Black and red dots indicate the position of the needle even during the treatment



Figure 4: Colour-coded clamps for arterial and venous needles

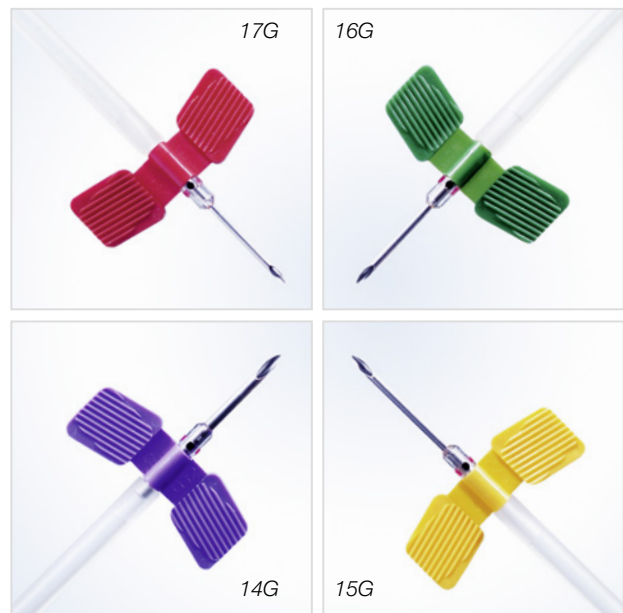


Figure 5: Wing colour indicates needle diameter

Fistula Needles sterilized by irradiation

Colour code of wing	Type	Needle (diameter x length)	Tubing length	Art. No.	Units per box
Standard					
 14 G	A401	2.0 x 20 mm	150 mm	507 740 1	50 x 1
	V401		150 mm	507 840 1	50 x 1
 15 G	A511	1.8 x 15 mm	150 mm	507 751 1	50 x 1
	V511		150 mm	507 851 1	50 x 1
	A501	1.8 x 20 mm	150 mm	507 750 1	50 x 1
	V501		150 mm	507 850 1	50 x 1
	A551	1.8 x 25 mm	150 mm	507 755 1	50 x 1
	V551		150 mm	507 855 1	50 x 1
 16 G	A611	1.6 x 15 mm	150 mm	507 761 1	50 x 1
	V611		150 mm	507 861 1	50 x 1
	A601	1.6 x 20 mm	150 mm	507 760 1	50 x 1
	V601		150 mm	507 860 1	50 x 1
	A651	1.6 x 25 mm	150 mm	507 765 1	50 x 1
	V651		150 mm	507 865 1	50 x 1
 17 G	A711	1.5 x 15 mm	150 mm	507 771 1	50 x 1
	V711		150 mm	507 871 1	50 x 1
	A701	1.5 x 20 mm	150 mm	507 770 1	50 x 1
	V701		150 mm	507 870 1	50 x 1
AV Sets (arterial and venous needle)					
 15 G	AV501	1.8 x 20 mm	150 mm	507 950 1	40 x 1
	AV552	1.8 x 25 mm	300 mm	507 655 1	40 x 1
 16 G	AV601	1.6 x 20 mm	150 mm	507 960 1	40 x 1
	AV652	1.6 x 25 mm	300 mm	507 665 1	40 x 1
 17 G	AV701	1.5 x 20 mm	150 mm	507 970 1	40 x 1
	AV752	1.5 x 25 mm	300 mm	507 676 1	40 x 1
Single needle					
 15 G	SN500	1.8 x 20 mm	100 mm	508 150 1	50 x 1
	SN550	1.8 x 25 mm	100 mm	508 155 1	50 x 1
 16 G	SN600	1.6 x 20 mm	100 mm	508 160 1	50 x 1
	SN650	1.6 x 25 mm	100 mm	508 165 1	50 x 1
 17 G	SN700	1.5 x 20 mm	100 mm	508 170 1	50 x 1

Coated with silicone, DEHP-free, latex-free, rotating wing

A: arterial needle, with back eye, red clamp
V: venous needle, without back eye, blue clamp
AV: arterial and venous set
SN: single needle, red and blue clamp

Fistula Needles sterilized with EtO (option 1)

Colour code of wing	Type	Needle (diameter × length)	Tubing length	Art. No.	Units per box
Standard					
 14 G	A	2.0 × 25 mm	150 mm	508244 1	50 × 1
	V		150 mm	508257 1	50 × 1
	A		300 mm	508249 1	50 × 1
	V		300 mm	508262 1	50 × 1
 15 G	A	1.8 × 25 mm	150 mm	508862 1	50 × 1
	V		150 mm	508863 1	50 × 1
	A		300 mm	508250 1	50 × 1
	V		300 mm	508263 1	50 × 1
 16 G	A	1.6 × 25 mm	150 mm	508864 1	50 × 1
	V		150 mm	508865 1	50 × 1
	A		300 mm	508251 1	50 × 1
	V		300 mm	508264 1	50 × 1
 17 G	A	1.5 × 25 mm	150 mm	508866 1	50 × 1
	V		150 mm	508867 1	50 × 1
	A		300 mm	508252 1	50 × 1
	V		300 mm	508265 1	50 × 1
Single needle					
 15 G	SN	1.8 × 20 mm	100 mm	508293 1	50 × 1
 16 G	SN	1.6 × 20 mm	100 mm	508294 1	50 × 1

Coated with silicone, latex-free, rotating wing

- A: arterial needle with back eye, red clamp
V: venous needle without back eye, blue clamp
SN: single needle, red and blue clamp
EtO: Ethylene Oxide

Fistula Needles sterilized with EtO (option 2)

Colour code of wing	Type	Needle (diameter x length)	Tubing length	Art. No.	Units per box
Standard					
 14 G	A	2.0 x 25 mm	150 mm	F00 004 741	100 x 1
	V		150 mm	F00 004 742	100 x 1
 15 G	A	1.8 x 25 mm	150 mm	F00 004 743	100 x 1
	V		150 mm	F00 004 744	100 x 1
	A		300 mm	F00 004 745	100 x 1
	V		300 mm	F00 004 746	100 x 1
 16 G	A	1.6 x 25 mm	150 mm	F00 004 747	100 x 1
	V		150 mm	F00 004 748	100 x 1
	A		300 mm	F00 004 749	100 x 1
	V		300 mm	F00 004 750	100 x 1
 17 G	A	1.5 x 25 mm	150 mm	F00 004 751	100 x 1
	V		150 mm	F00 004 752	100 x 1

Coated with silicone, DEHP-free, latex-free, rotating wing

A: arterial needle with back eye, red clamp
V: venous needle without back eye, blue clamp
EtO: Ethylene Oxide

Safety Fistula Needles

Helping to prevent stick injuries

The Facts

- More than 3 million needlestick injuries are estimated to occur worldwide annually¹
- These injuries present a substantial risk of infection with bloodborne viruses such as HBV, HCV and HIV²
- Up to 85 % of these injuries could be prevented with the use of protective and safe instruments²

Legislation is tightening:

- European Council Directive resolution on protecting healthcare workers from blood-borne infections due to needlestick injuries (2010)³ specifies the adoption of medical technology incorporating sharp protection mechanisms
- Technical guidelines “Biological agents in health care and welfare facilities” (“Biologische Arbeitsstoffe in Gesundheitswesen und in der Wohlfahrtspflege” – TRBA 250) in Germany define the use of safety needles for infectious patients
- U.S.A.: Needlestick Safety and Prevention Act (2001) and the Occupational Safety and Health Administration (OSHA) proclamation mandate the use of safety needles

Effective prevention

- Integrated safety feature (permanent locking) reduces the risk of accidental needle puncture and leads to improved safety in the workplace
- Following the European Council Directive
- Reduced risk of needlestick injury resulting in potential cost savings

Easy, user-friendly application

- High precision and low-pain puncture as well as fast and safe needle withdrawal
- Compatible with current clinical procedures
- Textured wings provide a secure grip

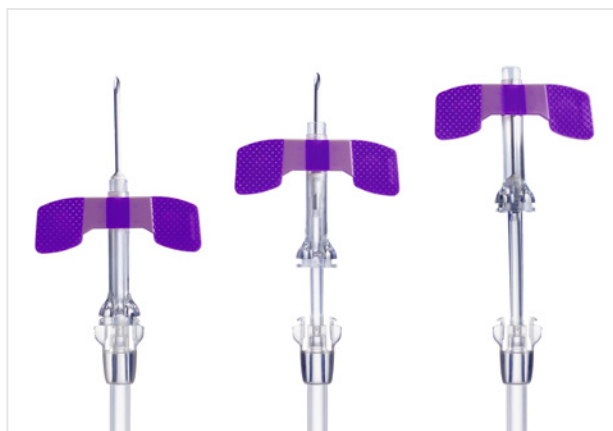
Reliable activation

- Pre-removal activation: Immediate shielding of the needle upon withdrawal from the fistula
- Once locked, it stays locked: To protect the user from needlestick injuries after use, the tip of the needle is pulled completely into a wing protector and then locked to prevent accidental loosening

Recommended Use of Fresenius Medical Care Safety Needles*:

After removal from the package, check to ensure that the wing protector is securely locked in the stopper before starting the treatment.

- Before starting the treatment, remove safety needle from packaging and check to ensure that the wing protector is securely locked in the stopper.
- At the end of the treatment, close the clamp, apply pressure to the puncture site and, using the same hand, hold down the wing protector.
- Release the lock by pressing both sides of the stopper, sliding the stopper back until it becomes fixed with an audible or tactile click.
- Once the wing protector is fully engaged, dispose of the protected needle in an approved sharps container.



*For complete information, please refer to the instructions for use provided with each package of Fresenius Medical Care safety needles.

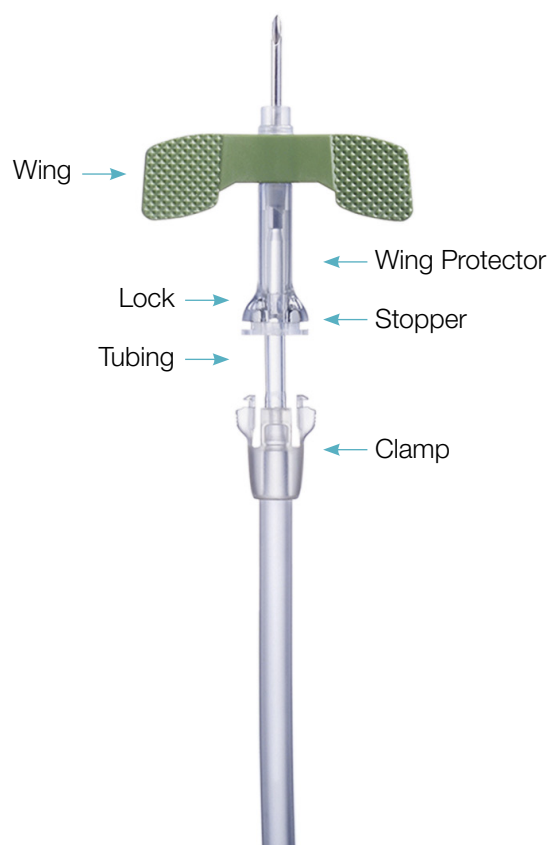
Product Range

Safety Fistula Needles

Colour code of wing	Type	Needle (diameter × length)	Tubing length	Art. No.	Units per box
Standard					
 14 G	S-14GA-R25L-R	2.0 × 25 mm	300 mm	507 832 1	50 × 1
	S-14GV-R25L-R		300 mm	507 836 1	50 × 1
 15 G	S-15GA-R25-R	1.8 × 25 mm	150 mm	507 722 1	50 × 1
	S-15GV-R25-R		150 mm	507 725 1	50 × 1
	S-15GA-R25L-R		300 mm	507 833 1	50 × 1
	S-15GV-R25L-R		300 mm	507 837 1	50 × 1
 16 G	S-16GA-R25-R	1.6 × 25 mm	150 mm	507 723 1	50 × 1
	S-16GV-R25-R		150 mm	507 726 1	50 × 1
	S-16GA-R25L-R		300 mm	507 834 1	50 × 1
	S-16GV-R25L-R		300 mm	507 838 1	50 × 1
 17 G	S-17GA-R25-R	1.5 × 25 mm	150 mm	507 724 1	50 × 1
	S-17GV-R25-R		150 mm	507 727 1	50 × 1
	S-17GA-R25L-R		300 mm	507 835 1	50 × 1
	S-17GV-R25L-R		300 mm	507 839 1	50 × 1

Sterilized by irradiation, coated with silicone, latex-free, rotating wing

- A: arterial needle with back eye, red clamp
V: venous needle, without back eye, blue clamp
L: long tubing



Buttonhole Needles

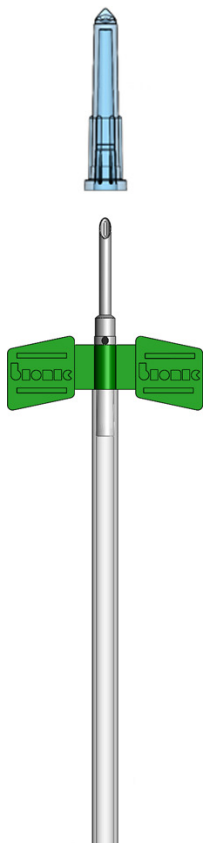
Colour code of wing	Type	Needle (diameter × length)	Tubing length	Art. No.	Units per box
Standard					
 15 G	A 846-BH S	1.8 × 20 mm	300 mm	TMD000561	50 × 1
	V 846-BH S	1.8 × 20 mm	300 mm	TMD000601	50 × 1
	A 856-BH S	1.8 × 25 mm	300 mm	TMD000571	50 × 1
	V 856-BH S	1.8 × 25 mm	300 mm	TMD000611	50 × 1
 16 G	A 646-BH S	1.6 × 20 mm	300 mm	TMD000541	50 × 1
	V 646-BH S	1.6 × 20 mm	300 mm	TMD000581	50 × 1
	A 656-BH S	1.6 × 25 mm	300 mm	TMD000551	50 × 1
	V 656-BH S	1.6 × 25 mm	300 mm	TMD000591	50 × 1

Sterilized by irradiation, coated with silicone, DEHP-free, latex-free, rotating wing, scab remover cap

- A:

arterial needle, with back eye, red clamp
- V:

venous needle, without back eye, blue clamp



Special needles for the Buttonhole puncturing technique with an integrated scab remover
 Manufacturer: Bionic Medizintechnik GmbH, Max-Planck-Str. 21, 61381 Friedrichsdorf, Germany, Tel. +49 (0) 6172-75760, www.bionic-jms.com

Importance of optimal needle size

In haemodialysis, solute clearance depends, among other factors, upon the effective blood flow (Q_B) passing through the dialyser. A high extracorporeal Q_B results in high dialysis efficacy Kt/V .⁴

Particularly in HighVolumeHDF® (postdilution haemo-diafiltration) a high blood flow is important to obtain adequate substitution volumes and subsequently a high middle molecule clearance.^{5,6}

Prerequisite to obtain required Q_B is a needle with adequate size. Figure 6 summarizes roughly recommendations for needle size in order to obtain the indicated blood flow rates.

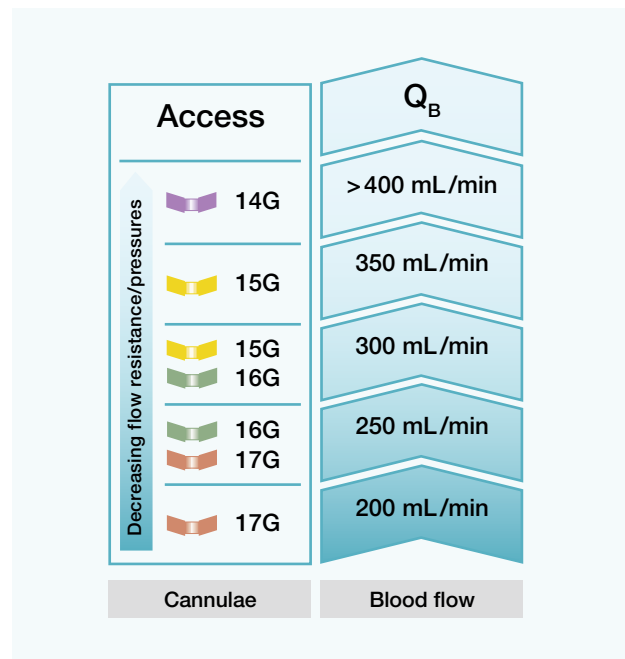


Figure 6: Gernally Recommended size of fistula needle in relation to the desired blood flow rate^{7,8}

The larger the diameter of the needle, the higher the blood flow at constant arterial and venous pressure.

For example, with a arterial pressure of -200 mmHg, a bigger needle diameter facilitates a significantly higher Q_B . (Figure 7).

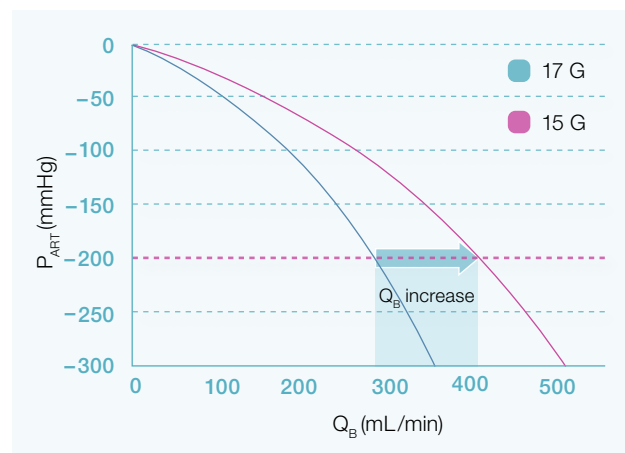


Figure 7: Switching to a bigger needle increases blood flow at constant arterial pressure

Unpublished to data on file: Fresenius Medical Care

Literature

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- 4 Ward: Blood Flow Rate: An Important Determinant of Urea Clearance and Delivered Kt/V. *Adv Ren Replace Ther* 1999 Jul;6(3): following 293.
- 5 Marcelli D, Kopperschmidt P, Bayh I, Jirka T, Merello JI, Ponce P, Ladanyi E, Di Benedetto A, Dovc-Dimec R, Rosenberger J, Stuard S, Scholz C, Grassmann A, Canaud B: Modifiable factors associated with achievement of high-volume post-dilution hemodiafiltration: results from an international study. *Int J Artif Organs*. 2015 May; 38(5):244-50.
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